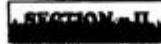


Attempt this Paper on Separate Answer Sheet provided.



Part-2 (SUBJECTIVE)

Q.2. Write short answers to each of the followings. (2x10)

- Describe Bartlett's test for homogeneity of variances.
- What are the assumptions in the application of χ^2 -test of goodness-of-fit?
- What are the properties of the sampling distribution of b , the estimate of β ?
- What is the basic difference between simple regression and multiple regression?
- What is the difference between $R_{1.23}$ and $r_{12.3}$?
- What is meant by Analysis of Variance?
- Why we use the LSD test in analysis of variance?
- What is the use of Duncan's multiple range test?
- What are disadvantages of completely randomized design?
- What are the basic principles of experimental designs?

Q.3. The weights of a random sample of 10 boxes of a particular brand of cereal are 14.2, 13.7, 14.1, 14.3, 14.1, 13.8, 14.4, 14.8, 13.9 and 14.3. Test the hypothesis that $H_0: \sigma^2 = 0.02$ against the alternative $H_1: \sigma^2 < 0.02$, using a 0.01 level of significance. (06)

Q.4. Three independent samples gave the following results: (06)

Size	Observations
5	34, 40, 47, 60, 84.
9	40, 59, 60, 67, 86, 92, 95, 98, 108.
3	46, 93, 100.

Use Bartlett's test to test the hypothesis of equal variances. Let $\alpha = 0.05$.

Q.5. An instructor of mathematics wished to determine the relationship of grades on a final examination to grades on two quizzes given during the semester. Calling X_1 , X_2 and X_3 the grades of a student on the first quiz, second quiz and final examination respectively, he made the following computations for a total of 120 students. (06)

$$\bar{X}_1 = 6.8 \quad S_1 = 1.0 \quad r_{12} = 0.60$$

$$\bar{X}_2 = 7.0 \quad S_2 = 0.8 \quad r_{13} = 0.70$$

$$\bar{X}_3 = 74 \quad S_3 = 9.0 \quad r_{23} = 0.65$$

Find the least-squares regression equation of X_3 on X_1 and X_2 .

P.T.O.

Q.6. In a feeding experiment of some animals, the following results were obtained, the numbers in the table being the gains in weight in pounds. The animals were in groups of 3 each. (06)

Groups	I	II	III	IV
A	7.0	16.0	10.5	13.5
Rations B	14.0	15.5	15.0	21.0
C	8.5	16.5	9.5	13.5

Test the hypothesis of no difference in rations, at $\alpha = 0.05$.

Q.7. Carry out the analysis of variance for the following Latin square. (06)

V1 (2.3)	V2 (3.0)	V3 (3.3)	V4 (2.5)
V2 (3.1)	V3 (4.1)	V4 (2.4)	V1 (2.4)
V3 (4.3)	V4 (2.5)	V1 (2.1)	V2 (2.9)
V4 (2.6)	V1 (2.0)	V2 (2.4)	V3 (4.4)

Q Describe Bartlett's test of homogeneity of variances?

The Bartlett's test homogeneity is

$$U = 2.30269$$

question 2(1)

$$q = (n-k) \log s_p^2 - \sum_{i=1}^k (n_i - 1) \log s_i^2$$

$$C = 1 + \frac{1}{3(k-1)} \left(\sum_{i=1}^k \frac{1}{n_i - 1} - \frac{1}{n-k} \right)$$

The value of q is large when the sample variances s_i^2 differ greatly. q is equal to zero when all s_i^2 are equal. Therefore we reject H_0 when calculated value of U is greater than or equal to tabulated value of χ^2 .

7. Chi-square distribution has one parameter.

Assumptions of Chi-Square Distribution.

1. The parent population is normal.
2. The sample is a random sample, so that the observations are independently distributed. It is most common assumption.

Applications of Chi-Square Distribution.

1. It is used as test of goodness of fit.
2. It is used to obtain confidence limits for the population variance.
3. It is used to find the association and independence between the attributes.
4. It is used to test the equality of several correlation coefficients.
5. It is used for the purpose of sampling technique.
6. It is used to test the equality of k variances.

question2(2)



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question 2(4)

The major difference between them is that while simple regression establishes the relationship between one dependent variable and one independent variable, multiple regression establishes the relationship between one dependent variable and more than one/ multiple independent variables.

question 2(5)

The partial correlation co-efficient varies between -1 and +1. The partial correlation formula is as follows:

Multiple correlation studies the effect of all the independent variables simultaneously on a dependent variable.

The value of the coefficient of multiple correlation lies between zero and one.



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question 2(6)

ANOVA stands for Analysis of Variance. It's a statistical test that was developed by Ronald Fisher in 1918 and has been in use ever since. Put simply, **ANOVA tells you if there are any statistical differences between the means of three or more independent groups.** One-way ANOVA is the most basic form.

* The Least Significant Difference Test:-

The Least square test is used in the context of the analysis of Variance When F-ratio suggest rejection of the null hypothesis H_0 that is when difference between population mean is significant.

$$LSD = t_{\alpha/2}(v) \sqrt{\frac{2(MSE)}{r}}$$

question 2(7)

all hypothesis." After obtaining the value of LSD we arrange the sample means in ascending order of magnitude and draw a line under any pair of adjacent means (or set of means) that are Not Significantly Different. The Significantly different pairs are immediately observed from their presentation. LSD test is a special testing procedure for pair wise comparison of group means.

question 2(8)

Duncan's Multiple Range Test.

This test which has different significant ranges, based on Studentized-range statistic $q(p, v)$ where p is the number of steps between two means being compared, but the values of $q(p, v)$ for appropriate values of p and v , the degrees of freedom of error mean square in the ANOVA, obtained from Duncan's table for significant ranges. It is

noted by $R_p = q_\alpha(p, v) \sqrt{\frac{(MSE)}{r}}$ where $p = 2, 3, \dots, k$

replications may be used.

- (v) The design is applicable only to a small number of treatments
- (vi) There is possibility of entering the whole of the variation among the experimental units into the experimental error as the randomization is not restricted in any direction.

Example 23.1. An experiment was conducted to compare the yields of three varieties of potatoes. Each variety was assigned at random to equal-size plots, four times. The yields were as follows:

Variety		
A	B	C

question 2(9) Disadvantage

discuss in the following

question 2(10)

23.2 BASIC PRINCIPLES OF EXPERIMENTAL DESIGNS

The basic principles of experimental designs are *randomization*, *replication* and *local control*. These principles make a valid test of significance possible. Each of them is described briefly in the following subsections.

R/C/B

(what is randomization?
purpose of randomization?)

How randomization occurs

The first principle of an experimental

(vi) **Conclusion.** Since the computed value does not fall in the critical region, so we accept H_0 .

Q.17.12 (a) (i) We state our null and alternative hypotheses as

$$H_0 : \sigma^2 = 0.02 \text{ and } H_1 : \sigma^2 < 0.02.$$

(ii) The level of significance is set at $\alpha = 0.01$.

(iii) The test-statistic is $\chi^2 = \frac{n S^2}{\sigma^2}$,

which has a χ^2 -distribution with $(n-1)$ degrees of freedom.

(iv) The critical region is $\chi^2 \leq \chi^2_{1-\alpha, (n)} = 2.09$

(v) Computations. $\sum X = 14.2 + 13.7 + \dots + 14.3 = 141.6$

question 3

$$\begin{aligned}\sum X^2 &= (14.2)^2 + (13.7)^2 + \dots + (14.3)^2 \\ &= 2005.98\end{aligned}$$

$$\begin{aligned}\therefore \sum (X - \bar{X})^2 &= \sum X^2 - \frac{(\sum X)^2}{n} \\ &= 2005.98 - \frac{(141.6)^2}{10} = 0.924\end{aligned}$$

$$\text{Now } \chi^2 = \frac{n S^2}{\sigma_0^2} = \frac{\sum (X - \bar{X})^2}{\sigma_0^2} = \frac{0.924}{0.02} = 46.2$$

(vi) **Conclusion.** Since our computed value of $\chi^2 = 46.2$ does not fall in the critical region, we therefore accept H_0 .

Question 17.14(b)

① Hypothesis:-

$$H_0: \sigma_1^2 = \sigma_2^2 = \sigma_3^2$$

H_1 : Not all variance are equal

② L.O.S:- $\alpha = 0.05$

③ Test statistic:-

$$U = 2.3026 \frac{q}{c}$$

$$q = (n-k) \log s_p^2 = \sum_{i=1}^k (n_i - 1) \log s_i^2$$

$$c = \frac{1}{3} \left(\frac{1}{k+1} \left(\sum_{i=1}^k \frac{1}{n_i - 1} - \frac{1}{n-k} \right) \right)$$

④ Calculations:-

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2
34	40	46	1156	1600	2116
40	59	93	1600	3481	8469
47	60	100	2209	3600	10000

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2
60	67	36	3600	4489	
84	86	7	7056	7396	
	92			8464	
	95			9025	
	98			9604	
	108			11664	
265	705	239	15621	59323	20785

question 4

$$S_1^2 = \frac{1}{n_1-1} \left(\sum X_1^2 - \frac{(\sum X_1)^2}{n_1} \right) = \frac{1}{5-1} \left(15621 - \frac{(265)^2}{5} \right) = 394$$

$$S_2^2 = \frac{1}{n_2-1} \left(\sum X_2^2 - \frac{(\sum X_2)^2}{n_2} \right) = \frac{1}{9-1} \left(59323 - \frac{(705)^2}{9} \right) = 512.25$$

$$S_3^2 = \frac{1}{n_3-1} \left(\sum X_3^2 - \frac{(\sum X_3)^2}{n_3} \right) = \frac{1}{3-1} \left(20785 - \frac{(239)^2}{3} \right) = \frac{772.33}{862.33}$$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2 + (n_3-1)S_3^2}{n_1 + n_2 + n_3 - 3}$$

$$= \frac{(5-1)394 + (9-1)512.25 + (3-1)862.33}{5+9+3-3} = \frac{1576 + 4098 + 1544.66}{14}$$

$$S_p^2 = 528.478$$

n_i	S_i^2	$\log S_i^2$	$(n_i-1) \log S_i^2$	$1/n_i-1$
5	394	2.595	10.38	0.25
9	512.25	2.709	21.672	0.125
3	862.33	2.935	5.8792	0.5
			37.827	0.875
			37.923	

$$q = \frac{(17-3)}{1} \log 528.478 - 37.923 = 38.122 - 37.827 = 0.199$$

$$C = 1 + \frac{1}{3(3-1)} \left(0.875 - \frac{1}{17-3} \right) =$$

question 4

$$C = 1.1339$$

$$U = 2.301 \frac{0.199}{1.1339} = 0.404$$

Critical region:-

$$U \geq \chi^2_{\alpha}(k-1)$$

$$0.404 \geq \chi^2_{0.05}(2)$$

$$0.404 \not\geq 5.99$$

Conclusion:- As calculated value does not lies in critical region so we will not rejected H_0 . We will accept H_0 .
17.15(a):-

Example 11.4 An instructor of mathematics wished to determine the relationship of grades in final examination to grades on two quizzes given during the semester. Calling X_1 , X_2 and X_3 the grades of a student on the first quiz, second quiz and final examination respectively, he made the following computations for a total of 120 students.

$$\begin{array}{lll} \bar{X}_1 = 6.8 & S_1 = 1.0 & r_{12} = 0.60 \\ \bar{X}_2 = 7.0 & S_2 = 0.8 & r_{13} = 0.70 \\ \bar{X}_3 = 74 & S_3 = 9.0 & r_{23} = 0.65 \end{array}$$

- Find the least-squares regression equation of X_3 on X_1 and X_2 .
- Estimate the final grades of two students who scored respectively (1) 9 and 7, and (2) 4 and 5 on the two quizzes.
- Compute $R_{3,12}$. (B.Sc. Eng. 1988)
- Since the standard deviations and linear correlation co-efficients are given, therefore estimated regression equation of X_3 on X_1 and X_2 is

$$\frac{X_3 - \bar{X}_3}{S_3} = \left(\frac{r_{13} - r_{12}r_{23}}{1 - r_{12}^2} \right) \left(\frac{X_1 - \bar{X}_1}{S_1} \right) + \left(\frac{r_{23} - r_{12}r_{13}}{1 - r_{12}^2} \right) \left(\frac{X_2 - \bar{X}_2}{S_2} \right)$$

Now $\frac{r_{13} - r_{12}r_{23}}{1 - r_{12}^2} = \frac{0.70 - (0.60)(0.65)}{1 - (0.60)^2} = \frac{0.31}{0.64}$, and

$$\frac{r_{23} - r_{12}r_{13}}{1 - r_{12}^2} = \frac{0.65 - (0.60)(0.70)}{1 - (0.60)^2} = \frac{0.23}{0.64}$$

question 5

Substituting these values, we get

$$\frac{X_3 - 74}{9.0} = \left(\frac{0.31}{0.64} \right) \left(\frac{X_1 - 6.8}{1.0} \right) + \left(\frac{0.23}{0.64} \right) \left(\frac{X_2 - 7.0}{0.8} \right)$$

or
$$\begin{aligned} X_3 - 74 &= 4.36 (X_1 - 6.8) + 4.04 (X_2 - 7.0) \\ &= 4.36 X_1 - 29.648 + 4.04 X_2 - 28.28 \end{aligned}$$

or $X_3 = 16.07 + 4.36 X_1 + 4.04 X_2$

is the desired least squares regression equation of X_3 on X_1 and X_2 .

b) Student 1: When $X_1 = 9$ and $X_2 = 7$, we get

$$\hat{X}_3 = 16.07 + 4.36(9) + 4.04(7) = 83.59 = 84$$

Student 2: When $X_1 = 4$ and $X_2 = 8$, we get

$$\hat{X}_3 = 16.07 + 4.36(4) + 4.04(8) = 65.83 = 66.$$

c) The co-efficient of multiple correlation $R_{3.12}$ is

$$\begin{aligned} R_{3.12} &= \sqrt{\frac{r_{31}^2 + r_{32}^2 - 2r_{12}r_{31}r_{32}}{1 - r_{12}^2}} \\ &= \sqrt{\frac{(0.70)^2 + (0.65)^2 - 2(0.60)(0.65)(0.70)}{1 - (0.60)^2}} \\ &= \sqrt{\frac{0.3665}{0.64}} = \sqrt{0.5727} = 0.757. \end{aligned}$$

question 5

Question No 20-19:-

(1) Hypothesis:-

Row H_0 : $\mu_A = \mu_B = \mu_C$

H_1 : Not all means are equal

Column

H_0 : $\mu_1 = \mu_2 = \mu_3 = \mu_4$

H_1 : Not all means are equal

(2) L.O.S:-

$$\alpha = 0.05$$

(3) Test statistic:-

$$F_1 = \frac{S_1^2}{S_3^2}$$

$$F_2 = \frac{S_2^2}{S_3^2}$$

(4) Calculation:-

	I	II	III	IV	$T_{i.}$	$T_{i.}^2$
Ratios A	7.0	16	10.5	13.5	47	2209
B	14.0	15.5	15.0	21	65.5	4290.25
C	8.5	16.5	9.5	13.5	48	2304
$T_{.j}$	29.5	48	35	48	160.5	8803.25
$T_{.j}^2$	870.25	2304	1225	2304	6703.25	
$\sum \sum X_{ij}^2$	317.25	768.5	425.5	805.5	2316.75	

Correction factor = C.F = $\frac{(T_{..})^2}{C \times R} = \frac{(160.5)^2}{4 \times 3} = 2146.68$

Total S-S = $2316.75 - 2146.68 = 170.07$

B/w S-S for ~~columns~~ $\rightarrow$ $\sum T_{.j}^2 - C.F = \frac{6703.25 - 2146.68}{3}$

$= 87.736$

B/w S-S for ~~rows~~ $\rightarrow$ $\sum T_{i.}^2 - C.F = \frac{8803.25 - 2146.68}{4}$

ratio

C

Question 6

$$= 28.202$$

$$= 170.07 - 87.736 - 54.1325$$

$$= 54.1325$$

$$\text{Error} = \text{Total S.S} - \text{Blw S.S row} - \text{Blw S.S column}$$

ANOVA Table:-

Source of Variation	d.f	S.S	M.S	F-ratio
Blw S.S for Column	$r-1 = 3-1 = 3$	87.73	$\frac{87.73}{3} = 29.24$	$F_1 = \frac{S_1^2}{S_2^2} = \frac{29.24}{7.0505} = 4.14$
Blw S.S for Rows	$c-1 = 4-1 = 3$	54.132	$\frac{54.132}{3} = 18.044$	
Error	$(r-1)(c-1) = 3 \times 3 = 9$	28.202	$\frac{28.202}{9} = 3.1336$	$F_2 = \frac{S_2^2}{S_3^2} = \frac{18.044}{3.1336} = 5.76$
Total	$rc-1 = 12-1 = 11$			

⑤ Critical region:-

1) We will reject H_0

$$F_c \geq F_{\alpha}(v_1, v_2)$$

$$4.14 > F_{0.05}(2, 4)$$

$$4.14 > 6.94$$

2) We will reject H_0

$$F_c \geq F_{\alpha}(v_1, v_2)$$

$$3.83 > F_{0.05}(3, 4)$$

$$3.83 > 6.59$$

⑥ Conclusion:-

As calculated value of F both does not lies in critical region so we will not reject H_0 we will accept H_0

Question No 23.22

2

question 7

① Hypothesis:-

$$H_0: T_1 = T_2 = T_3 = T_4$$

H_1 : The effect all treatment is not effective

② L.O.S:-

$$\alpha = 0.05$$

③ Test statistic

$$F = \frac{S_e^2}{S_o^2}$$

Row $\rightarrow i$

column $\rightarrow j$

Treatment $\rightarrow b$

④ Calculation:-

		Columns					
		1	2	3	4	R_i	R_i^2
Rows	1	2.3	3.0	3.3	2.5	11.1	123.21
	2	3.1	4.1	2.4	2.4	12	144
	3	4.3	2.5	2.1	2.9	11.8	139.24
	4	2.6	2.0	2.4	4.4	11.4	129.96
C_j		12.3	11.6	10.2	12.2	46.3	536.41
C_j^2		151.29	134.56	104.04	148.84	538.73	
$\sum Y_{ijk}^2$		40.15	36.06	26.82	39.78	142.81	

	Treatment				Total
	V ₁	V ₂	V ₃	V ₄	
$\bar{T}_k$	8.8	11.4	16.1	10	46.3
T_k^2	77.44	129.96	259.21	100	566.61

$$C.F = \frac{(\sum T_k)^2}{K^2} = \frac{(46.3)^2}{16} = 133.40$$

$$\text{Total S.S (Syy)} = \sum \sum Y_{ij}^2 - C.F = 142.81 - 133.40 = 8.83$$

$$\text{Column S.S } C_{yy} = \sum \frac{C_j^2}{K} - C.F = \frac{538.73}{4} - 133.98 = 0.782$$

$$\text{Row S.S } (R_{yy}) = \sum \frac{R_i^2}{K} - C.F = \frac{536.41}{4} - 133.98 = 0.1225$$

$$\text{Treatment S.S } (T_{yy}) = \sum \frac{T_k^2}{K} - C.F = \frac{566.61}{4} - 133.98 = 7.6725$$

$$\text{Error S.S } (E_{yy}) = S_{yy} - C_{yy} - R_{yy} - T_{yy} = 8.83 - 0.762 - 0.1225 - 7.67 = 0.3325$$

ANOVA Table

Source of Variation	d.f	S.S	M.S	F _{calc}
Rows	K-1 = 3	0.1225	0.0408	
Columns	K-1 = 3	0.7025	0.234	
Treatment	K-1 = 3	7.6725	2.5575	
Error	K-1 = 6	0.3325	0.0554	
Total	15	8.83		

15 - 9 = 6

$$F_1 = \frac{0.0908}{0.02216} = 4.11$$

$$F = \frac{s_t^2}{s_e^2} = \frac{2.5575}{0.0554}$$

$$F = 46.156$$

⑤ Critical Region We will reject H_0 if

$$F_c \geq F_{\alpha}(v_1, v_2)$$

$$F_c \geq F_{0.05}(3, 6)$$

$$45.53 \geq 4.76$$

⑥ Conclusion - As lie in rejection region so the effect of all treatment are not equal.